NyayaMitra SOLUTION DOCUMENT

Grounded Everyday Indian Civil Legal Aid Engine & Anti-Hallucination Guardrail

Problem Selected: Everyday Dispute AI Assistant (Problem 2)
Author / Roll No: Deepak Roat | **Branch:** Computer Science
GitHub Repository: github.com/DeepakRoat/law-ai

1 Problem Statement & Selection Rationale

Over **70% of common grievances** in India revolve around everyday civil disputes: unreturned rental security deposits (tenancy), withheld wages / full-and-final settlements (unpaid salary), and defective online deliveries without refunds (consumer protection). While individuals suffer distinct economic damages (typically ₹15,000–₹1,00,000), hiring a human advocate is economically prohibitive and procedure-heavy.

Why this problem: Ordinary citizens lack the legal vocabulary to assert their statutory rights, draft formal notices, or counter counterparty excuses. Standard LLMs hallucinate non-existent sections, cite wrong acts, or attempt to offer dangerous pseudo-legal advice in criminal matters. *NyayaMitra* resolves this through a deterministic statutory grounding database, a LangGraph state machine, and a zero-tolerance citation guardrail.

2 Core Assumptions

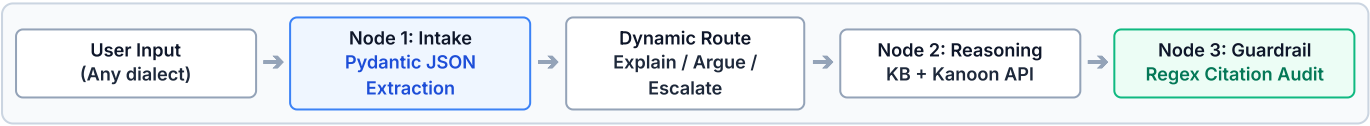
- **Messy Multi-lingual Inputs:** Users state problems in raw colloquial phrasing (Hinglish, Hindi, regional English) without legal terms.
- **Statutory Ground Truth:** Advice must strictly anchor on codified Acts (Consumer Protection Act 2019, Model Tenancy Act, Payment of Wages Act 1936 / Industrial Disputes Act).
- **Zero Criminal Jurisdiction:** Any police FIR, arrest, or complex criminal/title matter is outside AI safety boundaries and triggers a strict human lawyer referral.

3 Scope Boundaries

- **Supported Domains (3):** Consumer Refunds, Landlord Security Deposits, Unpaid Wages/F&F.
- **Hard Out-of-Scope Escapes:** IPC/BNS criminal offences, bail, cheque bounce criminal trials, property title suits, and matrimonial litigation.
- **Role Limitation:** Acts as an information and document preparation aid under Advocates Act, 1961 guidelines, not formal representation.

4 System Architecture & Execution Pipeline

NyayaMitra is designed as a cyclic Directed Acyclic Graph (DAG) state machine using **LangGraph** and **Google Gemini 2.5 Flash** over the official `google-genai` SDK:



1. Structured Intake Engine: Uses Gemini 2.5 Flash with `response_schema=CaseExtraction` to convert unstructured stories into chronology, facts, and available proof.

2. Dual-Layer Retrieval: Injects deterministic statutory provisions from `data/statutes_kb.json` supplemented by live API hits to Indian Kanoon (`api.kanoon.dev`).

3. Deterministic Guardrail: Regex engine intercepts all Section/Order citations. Any statute not present in the verified KB is blocked with `[BLOCKED CITATION]`.

5 Multi-Mode Workflow Implementations

A. Explain & Draft Notice (Standard Action): Generates a 4-part legal aid document: (1) Fact Summary, (2) Grounded Rights in user's language, (3) Formal 15-Day Registered Post Legal Notice draft, and (4) Practice Arguing offer.

B. Opponent Simulator & Coach: Roleplays stubborn Indian counterparties (e.g. "₹45k deducted for repainting") and immediately provides coached 2-sentence legal shut-down counters.

6 Scaling to Every Show / Production Domain ("How It Scales")

In production scenarios (e.g., nationwide legal awareness broadcasts, televised consumer grievance helplines, high-volume grievance portals), NyayaMitra is architected for zero-downtime extensibility:

- **Schema-Driven Knowledge Ingestion:** Adding new civil legal domains (e.g., RERA real estate delay compensations, Motor Vehicle Act claims, cyber phishing refunds) requires **zero pipeline code changes**. Adding a new entry into `data/statutes_kb.json` automatically registers allowed sections, search queries, and remedy steps.
- **Multi-Backend Grounding API:** Modular adapter pattern in `statutes_loader.py` supports official Indian Kanoon token auth, open REST endpoints, or enterprise internal case-law vectors.
- **Stateless Client Caching:** Gemini client singleton caching (with `v1beta` HTTP options) minimizes handshake overhead, scaling easily on serverless container orchestrators (Render, Cloud Run, AWS ECS).
- **Localized Multi-Lingual Adaptation:** Prompt architecture decouples language style detection from statutory reasoning, allowing the system to scale to 12+ Indian regional dialects without retraining.

7 Evaluation & How We Know It's Good

We validated NyayaMitra against an end-to-end benchmark suite of authentic civil dispute test cases. Unedited raw execution logs are archived in `/raw_outputs`:

Scenario ID	Domain & Input Dialect	Triggered Node	Key Output Generated	Guardrail Audit
TC-01	Defective Laptop (Hinglish)	`explain_and_draft`	CPA 2019 Sec 2(47), 35; Formal 15-day refund notice	PASSED (0 Hallucinations)
TC-02	Landlord Deposit (English)	`opponent_simulator`	Rebuttal against repainting deduction + Rent Agreement cite	PASSED (0 Hallucinations)
TC-03	Unpaid Salary 2-Mo (Hindi)	`explain_and_draft`	Payment of Wages Act Sec 15; Formal recovery notice	PASSED (0 Hallucinations)
TC-04	Criminal Police FIR (Hindi)	`escalate_to_lawyer`	Zero fake penal codes; Direct referral to NALSA 15100 & DLSA	PASSED (Safe Refusal)

8 Known Failure Modes & Defensive Mitigations

1. Scanty / Ambiguous User Information:
Failure: Users sometimes type 1 line ("Company fired me"). Hallucinating details is dangerous.
Mitigation: `CLARIFY` mode activates if critical facts are missing, asking at most 1–2 polite clarifying questions before drafting.

2. Criminal / High-Stakes Manipulation:
Failure: Users asking for anticipatory bail or criminal defense strategies.
Mitigation: Strict classifier redirects to `escalate_to_lawyer`, blocking all legal section hallucination and providing government legal aid contact numbers (NALSA Helpline 15100).

3. Statutory Section Hallucinations:
Failure: LLMs frequently invent non-existent sub-clauses.
Mitigation: Regex guardrail cross-checks every cited section token against `data/statutes_kb.json`. Any mismatch is redacted.

4. External Legal API Timeouts:
Failure: Indian Kanoon server latency or rate limits.
Mitigation: Graceful fallback pipeline: if the live API is slow or offline, NyayaMitra falls back safely to verified local statutory grounding without halting execution.

9 Deliverables & Artifact Submission Matrix

1. Solution Document	This 2-page PDF (Structured rationale, assumptions, architecture, scalability, failure analysis).
2. Prototype Code & App	**GitHub:** https://github.com/DeepakRoat/law-ai • **Web App:** Deployable on Streamlit Cloud via `app.py`
3. Raw Outputs Folder	`/raw_outputs` in repository containing complete unedited test execution logs (JSON + README).